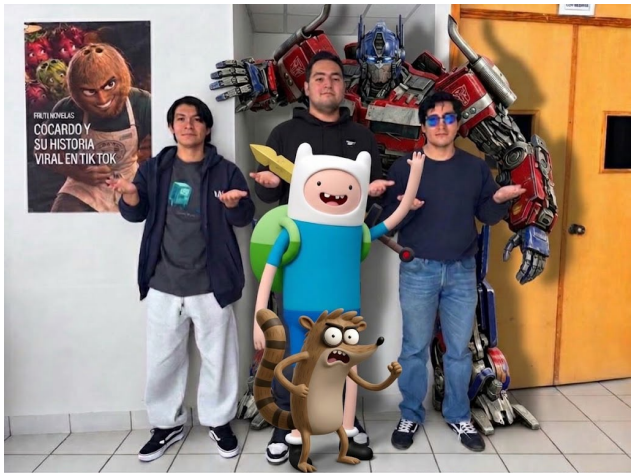
Base Julios GOD – Competitive Programming Reference



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1 Template

```cpp
#include <bits/stdc++.h>
using namespace std;
#define endl "\n"
#define all(x) x.begin(),x.end()
#define sz(x) (int)x.size()
#define SZ(x) (int)x.size()
#define rep(i,a,b) for(int i=a;i<b;i++)
typedef long double ld;
typedef long long ll;
void test(){

}
int main(){
    ios::sync_with_stdio(0); cin.tie(0);
    int t = 1; //cin >> t;
    while(t--) test();
}
```

2 Data structures

2.1 Segment tree

```cpp
#define M ((l + r) >> 1)
#define op(x, y) (x + y)
int st[4*N], values[N];
void build(int l, int r, int i) {
    if(l == r) {
        st[i] = values[l];
        return;
    }
    build(l, M, 2*i+1);
    build(M+1, r, 2*i+2);
    st[i] = op(st[2*i+1], st[2*i+2]); // #!
}
// Point update - range query
void update(int l, int r, int idx, int x, int i) {
    if(l == r) {
        st[i] += x;
        return;
    }
    if(idx <= M) update(l, M, idx, x, 2*i+1);
    else update(M+1, r, idx, x, 2*i+2);
    st[i] = op(st[2*i+1], st[2*i+2]);
}
int query(int l, int r, int a, int b, int i) {
    if(a > r || b < l) return 0;    // change for min/max
    if(a <= l && r <= b) return st[i];
    return op(query(l, M, a, b, 2*i+1), query(M+1, r, a, b, 2*i+2));
}
// l = 0, r = n-1, i = 0
```

2.2 Segment tree - Range update, point query

```cpp
// Range update - point query
// same as classic segment tree, but comment this line #!
void update(int l, int r, int a, int b, int x, int i) {
    if (a > r || b < l) return;
    if (a <= l && r <= b) {
        st[i] += x;
        return;
```

```
    }
    update(l, M, a, b, x, 2*i+1);
    update(M+1, r, a, b, x, 2*i+2);
}

int query(int l, int r, int idx, int i) {
    if(l == r) return st[i];

    int sum;
    if(idx <= M) sum = query(l, M, idx, 2*i+1);
    else sum = query(M+1, r, idx, 2*i+2);
    return op(sum, st[i]);
}
```

2.3 Segment tree - Lazy propagation

```
#define M ((l + r) >> 1)
#define op(x, y) (x + y)
ll st[4*N], lazy[4*N], arr[N];
void build(int l, int r, int i) {
    lazy[i] = 0;
    if (l == r) {st[i] = arr[l]; return;}
    build(l, M, 2*i+1);
    build(M+1, r, 2*i+2);
    st[i] = op(st[2*i+1], st[2*i+2]);
}
void push(int l, int r, int i) {
    if (!lazy[i]) return;
    st[i] += (r-l+1) * lazy[i]; // change for min/max
    if (l != r) {
        lazy[2*i+1] += lazy[i];
        lazy[2*i+2] += lazy[i];
    }
    lazy[i] = 0;  // change for multiplication
}
void update(int l, int r, int a, int b, ll x, int i) {
    push(l, r, i);
    if (a > r || b < l) return;
    if (a <= l && r <= b) {
        lazy[i] += x;
        push(l, r, i);
        return;
    }
    update(l, M, a, b, x, 2*i+1);
    update(M+1, r, a, b, x, 2*i+2);
    st[i] = op(st[2*i+1], st[2*i+2]);
}
ll query(int l, int r, int a, int b, int i) {
    if (a > r || b < l) return 0; // change for min/max
    push(l, r, i);
    if (a <= l && r <= b) return st[i];
    return op(query(l, M, a, b, 2*i+1), query(M+1, r, a, b, 2*i+2));
}  // i=0, l=0, r=n-1, x=value, a,b=range query
```

2.4 Segment tree - Persistence

```
#define M ((l + r) >> 1)
struct Node{
    Node *left, *right;
    ll val;
    Node(ll x) : left(NULL), right(NULL), val(x) {}   // Add value
    Node(Node *l, Node *r) : left(l), right(r), val(0) { // Update values
        if(l) val += l->val;
        if(r) val += r->val;
    }
    Node(Node *root) : left(root->left), right(root->right), val(root->val) {} //
        Make copy
};
Node *build(int l, int r, vector<int> &values) {
    if(l == r) return new Node(values[l]);
    return new Node(build(l, M, values), build(M+1, r, values));
}
Node *update(Node *node, int l, int r, int idx, ll k) {
    if(l == r) return new Node(k);
    if(idx <= M) return new Node(update(node->left, l, M, idx, k), node->right);
    return new Node(node->left, update(node->right, M+1, r, idx, k));
}
ll query(Node *node, int l, int r, int a, int b) {
    if(a > r || b < l) return 0;
    if(a <= l && r <= b) return node->val;
    return query(node->left, l, M, a, b) + query(node->right, M+1, r, a, b);
}
Node *roots[N];
// 0 based indexing
// roots[copy++] = build(0, n-1);
// roots[copy++] = new Node(roots[--idx]);
// roots[--copy] = update(roots[--copy], 0, n-1, --idx, x);
// query(roots[--copy], 0, n-1, --a, --b)
```

2.5 Fenwick tree

```
struct FT{
    int n;
    vector<int> ft;
    FT(int _n) : n(_n), ft(_n+1) {}
    void add(int idx, int k) {
        for(; idx<=n; idx+=idx&-idx)
            ft[idx] += k;
    }
```

```
    int query(int idx) {
        int sum = 0;
        for(; idx>0; idx-=idx&-idx)
            sum += ft[idx];
        return sum;
    }
    int query(int l, int r) {
        return query(r) - query(l-1);
    }
    int lower_bound(int k) { // LOG = log2(n) + 1
        int sum = 0, idx = 0;
        for(int i=LOG-1; i>=0; i--) {
            if(idx + (1 << i) <= n && sum + ft[idx + (1 << i)] < k) {
                sum += ft[idx + (1 << i)];
                idx += 1 << i;
            }
        }
        return idx + 1;
    }
}; // 1-based indexing
```

2.6 DSU

```
struct DSU{
    int n;
    vector<int> parent, rank;
    DSU(int _n) : n(_n), parent(_n), rank(_n) {
        for(int i=0; i<n; i++) {
            parent[i] = i;
            rank[i] = 0;
        }
    }
    int find_set(int v) {
        if (v == parent[v]) return v;
        return parent[v] = find_set(parent[v]);
    }
    void union_sets(int a, int b) {
        a = find_set(a), b = find_set(b);
        if (a != b) {
            if (rank[a] < rank[b]) swap(a, b);
            parent[b] = a;
            if (rank[a] == rank[b]) rank[a]++;
        }
    }
    int components() {
        int cnt = 0;
        for(int i=0; i<n; i++) {
            if(find_set(i) == i) cnt++;
        }
        return cnt;
    }
};
```

2.7 SQRT decomposition

```
struct SQ{
    int n, b;
    vector<int> values, blocks;
    SQ(int _n) : n(_n), values(_n) {
        b = (int)sqrt(n) + 1;
        blocks = vector<int> (b);
    }
    // Basic update / query
    void operation(int l, int r, int k) {
        int bl = l/b, br = r/b;
        // operation lies in same block
        if(bl == br) {
            for(int i=l; i<=r; i++) {
                blocks[bl] -= values[i];
                values[i] += k;
                blocks[bl] += values[i];
            }
        }
        // operation on different blocks
        else {
            for(int i=l; i<(bl+1)*b; i++) {
                blocks[bl] -= values[i];
                values[i] += k;
                blocks[bl] += values[i];
            }
            for(int i=br*b; i<=r; i++) {
                blocks[br] -= values[i];
                values[i] += k;
                blocks[br] += values[i];
            }
            for(int i=(bl+1)*b; i<br; i++) {
                blocks[i] += k * b;
            }
        }
    }
};
```

2.8 Trie

```
struct Node{
    vector<Node*> ocu;
    bool flag;
    Node() : ocu(26), flag(false) {}
};
```

```
struct Trie{
    Node *root;
    Trie() : root(new Node()) {}
    void insert(string word) {
        Node *curr = root;
        for(auto &it : word) {
            if(!curr -> ocu[it - 'a'])
                curr -> ocu[it - 'a'] = new Node();
            curr = curr -> ocu[it - 'a'];
        }
        curr -> flag = true;
    }
    bool search(string word) {
        Node *curr = root;
        for(auto &it : word) {
            if(!curr -> ocu[it - 'a']) return false;
            curr = curr -> ocu[it - 'a'];
        }
        return curr -> flag;
    }
};
```

2.9 Trie XOR

```
struct Node{
    vector<Node*> ocu;
    Node() : ocu(2) {}
};
struct Trie {
    Node *root;
    Trie() : root(new Node()) { insert(0); }

    void insert(long long x) {
        Node *curr = root;
        for(int mask=63; mask>=0; mask--) {
            bool currBit = (x >> mask) & 1;
            if(!curr -> ocu[currBit])
                curr -> ocu[currBit] = new Node();
            curr = curr -> ocu[currBit];
        }
    }
    long long query(long long prefix) {
        Node *curr = root;
        long long res = 0;
        for(int mask=63; mask>=0; mask--) {
            bool currBit = (prefix >> mask) & 1;
            if(curr -> ocu[currBit ^ 1]) {
                res |= (1LL << mask);
                curr = curr -> ocu[currBit ^ 1];
            }
            else curr = curr -> ocu[currBit];
        }
        return res;
    }
};
```

2.10 Map custom hash

```
#include <ext/pb_ds/assoc_container.hpp>
using namespace __gnu_pbds;

struct custom_hash {
    static uint64_t splitmix64(uint64_t x) {
        x += 0x9e3779b97f4a7c15;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb;
        return x ^ (x >> 31);
    }
    size_t operator()(uint64_t x) const {
        static const uint64_t FIXED_RANDOM = chrono::steady_clock::now().
            time_since_epoch().count();
        return splitmix64(x + FIXED_RANDOM);
    }
};
gp_hash_table<int, int, custom_hash> freq;
```

2.11 Ordered set

```
#include<ext/pb_ds/assoc_container.hpp>
#include<ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
typedef tree<int,null_type,less<int>,rb_tree_tag,
    tree_order_statistics_node_update> ordered_set;
// find_by_order(i) -> iterator to ith element
// order_of_key(k) -> position (int) of lower_bound of k
```

2.12 Merge Sort Tree

```
#define M ((l + r) >> 1)
struct ST{
    int n;
    vector<vector<int>> st;
    ST(int _n) : n(_n), st(4*_n) {}

    void build(int l, int r, int i, vector<int> &values) {
        if(l == r) {
            st[i].pb(values[l]);
            return;
        }
```

```
        build(l, M, 2*i+1, values);
        build(M+1, r, 2*i+2, values);
        st[i].resize(st[2*i+1].size() + st[2*i+2].size());
        merge(st[2*i+1].begin(), st[2*i+1].end(), st[2*i+2].begin(), st[2*i+2].
            end(), st[i].begin());
    }
    int query(int l, int r, int a, int b, int k, int i) {
        if(a > r || b < l) return 0;
        if(a <= l && r <= b) return st[i].end() - upper_bound(st[i].begin(), st[i
            ].end(), k);
        return query(l, M, a, b, k, 2*i+1) + query(M+1, r, a, b, k, 2*i+2);
    }
};
```

2.13 Wavelet Tree

```
// indexed in 1
// from pointer to first element and to to end
// x and y The minimun element and y the max element
// If you need only one function or more erase the others
// If you need tu construct other function you only required to undertand the
    limit, this
// are the same
struct wavelet_tree{
  int lo, hi;
  wavelet_tree *l, *r;
  vector<int> b;
  wavelet_tree(int *from, int *to, int x, int y){
    lo = x, hi = y;
    if(lo == hi or from >= to) return;
    int mid = (lo+hi)/2;
    auto f = [mid](int x){ return x <= mid;};
    b.reserve(to-from+1);
    b.pb(0);
    for(auto it = from; it != to; it++)
      b.push_back(b.back() + f(*it));
    auto pivot = stable_partition(from, to, f);
    l = new wavelet_tree(from, pivot, lo, mid);
    r = new wavelet_tree(pivot, to, mid+1, hi);
  }
  //kth smallest element in [l, r]
  int kth(int l, int r, int k){
    if(l > r) return 0;
    if(lo == hi) return lo;
    int inLeft = b[r] - b[l-1];
    int lb = b[l-1];
    int rb = b[r];
    if(k <= inLeft) return this->l->kth(lb+1, rb , k);
    return this->r->kth(l-lb, r-rb, k-inLeft);
  }
  //count of nos in [l, r] Less than or equal to k
  int LTE(int l, int r, int k) {
    if(l > r or k < lo) return 0;
    if(hi <= k) return r - l + 1;
    int lb = b[l-1], rb = b[r];
    return this->l->LTE(lb+1, rb, k) + this->r->LTE(l-lb, r-rb, k);
  }
  //count of nos in [l, r] equal to k
  int count(int l, int r, int k) {
    if(l > r or k < lo or k > hi) return 0;
    if(lo == hi) return r - l + 1;
    int lb = b[l-1], rb = b[r], mid = (lo+hi)/2;
    if(k <= mid) return this->l->count(lb+1, rb, k);
    return this->r->count(l-lb, r-rb, k);
  }
};
```

2.14 Monotonic stack

```
int main() {
    ios::sync_with_stdio(0); cin.tie(0);
    int n; cin >> n;
    vector<int> v(n), l(n), r(n);
    for(auto &it : v) cin >> it;

    stack<pair<int, int>> st;
    for(int i=n-1; i>=0; i--) {
        while(!st.empty() && v[i] < st.top().first) {
            l[st.top().second] = i;
            st.pop();
        }
        st.push({v[i], i});
    }

    while(!st.empty()) l[st.top().second] = -1, st.pop();

    rep(i,0,n) {
        while(!st.empty() && v[i] <= st.top().first) {
            r[st.top().second] = i;
            st.pop();
        }
        st.push({v[i], i});
    }

    while(!st.empty()) r[st.top().second] = n, st.pop();

    vector<int> res(n);
    rep(i,0,n) {
```

```
        int curr = r[i] - l[i] - 1;
        res[curr-1] = max(res[curr-1], v[i]);
    }
    for(int i=n-2; i>=0; i--) {
        res[i] = max(res[i], res[i+1]);
    }

    for(auto &it : res) cout << it << " ";
    cout << endl;

    return 0;
}
```

2.15 Matrix operations

```
struct Matrix {
  ll a[N][N];
  Matrix() {memset(a,0,sizeof(a));}
  Matrix operator *(Matrix other) { // Product of a matrix
    Matrix product=Matrix();
        rep(i,0,N) rep(j,0,N) rep(k,0,N) {
            product.a[i][k]+=a[i][j]*other.a[j][k];
            product.a[i][k]%=MOD;
        }
    return product;
  }
};
Matrix expo_power(Matrix a, ll n) { // Matrix exponentiation
  Matrix res=Matrix();
    rep(i,0,N) res.a[i][i]=1; // Matriz identidad
  while(n){ // binpow
        if(n & 1) res = res * a;
        n >>= 1;
        a = a * a;
  }
  return res;
} // Ej. Matrix M=Matrix(); M.a[0][0]=1; M=M*M; Matrix res=expo_power(M,k);
```

2.16 Heavy Light Decomposition

```
int sz[N], in[N], out[N], timer, head[N], parent[N];
void dfs_sz(int u = 0, int p = -1) {
    sz[u] = 1;
    parent[u] = p;
    for(auto &v : adj[u]) {
        if(v == p) continue;
        dfs_sz(v, u);
        sz[u] += sz[v];
        if(adj[u][0] == p || sz[v] > sz[adj[u][0]]) {
            swap(v, adj[u][0]);
        }
    }
}
void dfs_hld(int u = 0, int p = -1) {
    in[u] = timer++;
    for(auto &v : adj[u]) {
        if(v == p) continue;
        head[v] = (adj[u][0] == v ? head[u] : v);
        dfs_hld(v, u);
    }
    out[u] = timer;
}
ll query_path(int u, int v) {
    ll res = 0;
    while(head[u] != head[v]) {
        if(in[u] < in[v]) swap(u, v);
        res += query(0, timer-1, in[head[u]], in[u], 0);
        u = parent[head[u]];
    }
    if(in[u] > in[v]) swap(u, v);
    res += query(0, timer-1, in[u]+1, in[v], 0);
    // in[u] (node) / in[u]+1 (edge)
    return res;
}
```

2.17 Convex Hull Trick

```
typedef ll tc;
struct Line{tc m,h;};
struct CHT { // for minimum (for maximum just change the sign of lines)
  vector<Line> c;
  int pos=0;
  tc in(Line a, Line b){
    tc x=b.h-a.h,y=a.m-b.m;
    return x/y+(x%y?!((x>0)^(y>0)):0); // ==ceil(x/y)
  }
  void add(tc m, tc h){ // m's should be non increasing
    Line l=(Line){m,h};
    if(c.size()&&m==c.back().m){
      l.h=min(h,c.back().h);c.pop_back();if(pos)pos--;
    }
    while(c.size()>1&&in(c.back(),l)<=in(c[c.size()-2],c.back())){
      c.pop_back();if(pos)pos--;
    }
    c.pb(l);
  }
  inline bool fbin(tc x, int m){return in(c[m],c[m+1])>x;}
  tc query(tc x){
    // O(log n) query:
```

```
    int s=0,e=c.size();
    while(e-s>1){int m=(s+e)/2;
      if(fbin(x,m-1))e=m;
      else s=m;
    }
    return c[s].m*x+c[s].h;
    // O(1) query (for ordered x's):
    while(pos>0&&fbin(x,pos-1))pos--;
    while(pos<c.size()-1&&!fbin(x,pos))pos++;
    return c[pos].m*x+c[pos].h;
  }
}; // y=mx+b -> cht.add(m, b) | cht.query(x)
```

2.18 Dynamic Convex Hull Trick

```
using ll = long long;
struct Line {
  mutable ll m, b, p;
  bool operator<(const Line& o) const { return m < o.m; }
  bool operator<(ll x) const { return p < x; }
};
struct LineContainer : multiset<Line, less<>> {
  // (for doubles, use inf = 1/.0, div(a,b) = a/b)
  static const ll inf = LLONG_MAX;
  ll div(ll a, ll b) { // floored division
    return a / b - ((a ^ b) < 0 && a % b); }
  bool isect(iterator x, iterator y) {
    if (y == end()) return x->p = inf, 0;
    if (x->m == y->m) x->p = x->b > y->b ? inf : -inf;
    else x->p = div(y->b - x->b, x->m - y->m);
    return x->p >= y->p;
  }
  void add(ll m, ll b) {
    auto z = insert({m, b, 0}), y = z++, x = y;
    while (isect(y, z)) z = erase(z);
    if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
    while ((y = x) != begin() && (--x)->p >= y->p)
      isect(x, erase(y));
  }
  ll query(ll x) {
    assert(!empty());
    auto l = *lower_bound(x);
    return l.m * x + l.b;
  }
}; // y=mx+b -> cht.add(m, b) | cht.query(x)
```

3 Graphs

3.1 Euler tour

```
vector<int> in(n+1), out(n+1), euler(n+1);
int timer = 0;
auto dfs = [&](auto self, int u, int p = -1) -> void {
    in[u] = ++timer;
    euler[timer] = u;
    for(auto &v : adj[u]) {
        if(v == p) continue;
        self(self, v, u);
    }
    out[u] = timer;
};
dfs(dfs, 0);
queries[i] = {in[u], out[u], k, i};
int x = values[euler[idx]];
```

3.2 Find bridges

```
int in[N], low[N], timer;
/* Articulation bridges */
void dfs(int u, int p = -1) {
    in[u] = low[u] = ++timer;
    for(auto &v : adj[u]) {
        if(v == p) continue;
        if(!in[v]) {
            dfs(v, u);
            low[u] = min(low[u], low[v]);
            if(low[v] > in[u]) IS_BRIDGE(U, V);
        }
        else low[u] = min(low[u], in[v]);
    }
}

/* Bridge tree */
// Use dfs to find bridges, but change
if(low[v] <= in[u]) dsu.union_sets(u, v);

// Iterate over the edges
for(auto [u, v] : edges) {
    u = dsu.find_set(u);
    v = dsu.find_set(v);
    if(u != v) {
        bridgeTree[u].pb(v);
        bridgeTree[v].pb(u);
    }
}

/* Articulation points */
// Use same code as finding bridges
if(low[v] >= in[u]) IS_ARTICULATION_POINT(U, V);
```

3.3 Dijkstra

```cpp
void dijkstra(int start, int n) {
    priority_queue<pair<ll, int>, vector<pair<ll,int>>, greater<pair<ll,int>>>
        pq;
    for(int i=0; i<n; i++) {
        dist[i] = 1e14;
    }

    dist[start] = 0;
    pq.push({0, start});

    while(!pq.empty()) {
        auto [d, u] = pq.top();
        pq.pop();

        if(d > dist[u]) continue;

        for(auto &[v, w] : adj[u]) {
            if(dist[u] + w < dist[v]) {
                dist[v] = dist[u] + w;
                pq.push({dist[v], v});
            }
        }
    }
}
```

3.4 Binary Lifting LCA

```cpp
int depth[N], up[N][LOG];
void dfs1(int u, int p) {
    up[u][0] = p;
    rep(i,1,LOG) {
        if(up[u][i-1] != -1)
            up[u][i] = up[up[u][i-1]][i-1];
        else
            up[u][i] = -1;
    }
    for(auto &v : adj[u]) {
        if(v == p) continue;
        depth[v] = depth[u] + 1;
        dfs1(v, u);
    }
}
int lift(int u, int k) {
    for(int i=LOG-1; i>=0; i--) {
        if(k & (1 << i)) {
            u = up[u][i];
            if(u == -1) return 0;
        }
    }
    return u;
}
int lca(int u, int v) {
    if(depth[u] < depth[v]) swap(u, v);
    u = lift(u, depth[u] - depth[v]);
    if(u == v) return u;

    for(int i=LOG-1; i>=0; i--) {
        if(up[u][i] != -1 && up[u][i] != up[v][i]) {
            u = up[u][i];
            v = up[v][i];
        }
    }
    return up[u][0] == -1 ? 0 : up[u][0];
}
```

3.5 Binary Lifting - Operations on path

```cpp
// op de los pesos de las aristas en el camino u-v (min, max, gcd, ...)
// Llamar dfs1(root, root, NEUTRO)
#define op(a,b) min(a,b)
const int NEUTRO = 1e9;
int depth[N], up[N][LOG], upm[N][LOG];
void dfs1(int u, int p, int w) {
    up[u][0] = p;
    upm[u][0] = w;
    rep(i,1,LOG) {
        up[u][i] = up[up[u][i-1]][i-1];
        upm[u][i] = op(upm[u][i-1], upm[up[u][i-1]][i-1]);
    }
    for(auto &[v, c] : adj[u]) {
        if(v == p) continue;
        depth[v] = depth[u] + 1;
        dfs1(v, u, c);
    }
}
int query(int u, int v) {
    int ans = NEUTRO;
    if(depth[u] < depth[v]) swap(u, v);
    for(int i=LOG-1; i>=0; i--) {
        if(depth[u] - (1 << i) >= depth[v]) {
            ans = op(ans, upm[u][i]);
            u = up[u][i];
        }
    }
    if(u == v) return ans;
    for(int i=LOG-1; i>=0; i--) {
```

```
        if(up[u][i] != up[v][i]) {
            ans = op(ans, op(upm[u][i], upm[v][i]));
            u = up[u][i];
            v = up[v][i];
        }
    }
    return op(ans, op(upm[u][0], upm[v][0]));
}
```

3.6 Centroid decomposition

```
void dfs(int u, int p) {
    sz[u] = 1;
    for(auto &v : adj[u]) {
        if(v == p || centroid[v]) continue;
        dfs(v, u);
        sz[u] += sz[v];
    }
}
int getCentroid(int u, int p, int n) {
    for(auto &v : adj[u]) {
        if(v == p || centroid[v]) continue;
        if(2 * sz[v] > n) return getCentroid(v, u, n);
    }
    return u;
}
void getCount(int u, int p, int depth) {
    if(depth > k) return;
    res += 0ll + cnt[k - depth];
    for(auto &v : adj[u]) {
        if(v == p || centroid[v]) continue;
        getCount(v, u, depth+1);
    }
}
void getPaths(int u, int p, int depth, int x) {
    if(depth > k) return;
    cnt[depth] += x;
    for(auto &v : adj[u]) {
        if(v == p || centroid[v]) continue;
        getPaths(v, u, depth+1, x);
    }
}
void decompose(int u, int p) {
    dfs(u, p);
    int c = getCentroid(u, p, sz[u]);
    centroid[c] = 1;
    cnt[0] = 1;

    for(auto &v : adj[c]) {
        if(centroid[v]) continue;
        getCount(v, c, 1);
        getPaths(v, c, 1, 1);
    }

    getPaths(c, p, 0, -1);

    for(auto &v : adj[c]) {
        if(v == c || centroid[v]) continue;
        decompose(v, c);
    }
}
```

3.7 Centroid decomposition - Fast

```
// Centroid decomposition: O(n log n), el arbol de centroides tiene
// profundidad O(log n). par[c] = padre de c en el arbol de centroides.
// Llamar decompose(1). Todo camino u-v pasa por el centroide que los separa.
int sz[N], par[N];
bool removed[N];
// tamano de los subarboles (ignorando nodos ya removidos)
int dfs(int u, int p) {
    sz[u] = 1;
    for(int v : adj[u]) if(v != p && !removed[v]) sz[u] += dfs(v, u);
    return sz[u];
}
// baja al hijo con mas de n/2 nodos hasta que no haya ninguno
int getCentroid(int u, int p, int n) {
    for(int v : adj[u]) if(v != p && !removed[v] && 2 * sz[v] > n) return
        getCentroid(v, u, n);
    return u;
}
void decompose(int u, int p = -1) {
    int c = getCentroid(u, -1, dfs(u, -1));
    removed[c] = 1, par[c] = p;
    // aqui va lo del problema, con c como raiz
    for(int v : adj[c]) if(!removed[v]) decompose(v, c);
}
```

3.8 Edmonds karp

```
const int N = 1e3+5;
vector<int> adj[N];
int parent[N], cap[N][N];

void add_edge(int u, int v, int c) {
    adj[u].pb(v);
    adj[v].pb(u);
    cap[u][v] += c;
}
```

```cpp
int bfs(int s, int t) {
    memset(parent, -1, sizeof(parent));
    parent[s] = -2;
    queue<pair<int, int>> q;
    q.push({s, (int)1e9});

    while(!q.empty()) {
        auto [u, f] = q.front();
        q.pop();

        for(auto &v : adj[u]) {
            if(parent[v] != -1 || !cap[u][v]) continue;

            parent[v] = u;
            int flow = min(cap[u][v], f);
            if(v == t) return flow;

            q.push({v, flow});
        }
    }
    return 0;
}
int maxFlow(int s, int t) {
    int mx = 0, flow;
    while(flow = bfs(s, t)) {
        mx += flow;
        int curr = t;
        while(curr != s) {
            int prev = parent[curr];
            cap[prev][curr] -= flow;
            cap[curr][prev] += flow;
            curr = prev;
        }
    }
    return mx;
}
```

3.9 Dinics

```cpp
struct Dinic {
    int nodes, src, dst;
    vector<int> dist, q, work;
    struct edge{int to, rev; ll f, cap;};
    vector<vector<edge>> g;
    Dinic(int x) : nodes(x), g(x), dist(x), q(x), work(x){}
    void add_edge(int s, int t, ll cap) {
        g[s].pb((edge) {t, SZ(g[t]), 0, cap});
        g[t].pb((edge) {s, SZ(g[s])-1, 0, 0});
    }
    bool dinic_bfs() {
        fill(all(dist), -1); dist[src]=0;
        int qt=0; q[qt++]=src;
        for(int qh=0; qh<qt; qh++) {
            int u=q[qh];
            rep(i, 0, SZ(g[u])) {
                edge &e = g[u][i]; int v=g[u][i].to;
                if(dist[v]<0&&e.f<e.cap) dist[v]=dist[u]+1, q[qt++]=v;
            }
        }
        return dist[dst]>=0;
    }
    ll dinic_dfs(int u, ll f) {
        if(u==dst) return f;
        for(int &i=work[u]; i<SZ(g[u]); i++) {
            edge &e=g[u][i];
            if(e.cap <= e.f) continue;
            int v=e.to;
            if(dist[v]==dist[u]+1) {
                ll df=dinic_dfs(v, min(f, e.cap-e.f));
                if(df > 0) {e.f+=df; g[v][e.rev].f-=df; return df;}
            }
        }
        return 0;
    }
    ll max_flow(int _src, int _dst) {
        src=_src, dst=_dst;
        ll result=0;
        while(dinic_bfs()) {
            fill(all(work), 0);
            while(ll delta=dinic_dfs(src, 1e12)) result+=delta;
        }
        return result;
    }
};

int main() {io
    int n,m,sl,el,s,t,u,v;
    cin>>n>>sl>>el;
    s=n+n, t=s+1;
    Dinic nf(n+n+2);
    rep(i,0,sl){
        cin>>u; u--;
        nf.add_edge(s,u,1);
    }
    rep(i,0,el){
        cin>>u; u--;
        nf.add_edge(u+n,t,1);
```

```
    }
    rep(i,0,n) nf.add_edge(i,i+n,1);
    cin>>m;
    rep(i,0,m){
        int u,v;
        cin>>u>>v;
        u--,v--;
        nf.add_edge(u+n,v,1);
    }
  cout<<nf.max_flow(s,t)<<endl;
    return 0;
}
```

3.10 Bipartite check

```
int color[N];
bool f = 1;
// memset(color, -1, sizeof(color));
bool dfs(int u = 0, int c = 0) {
    color[u] = c;
    for(auto &v : adj[u]) {
        if(color[v] == -1 && !dfs(v, 1-c)) return false;
        else if(color[v] == c) return false;
    }
}
```

3.11 Has cycle?

```
int color[N];
// directed graph
bool dfs(int u, int p = -1) {
    color[u] = 1;
    for(auto &v : adj[u]) {
        if(!color[v] && dfs(v, u)) return true;
        if(color[v] == 1) return true;
    }
    color[u] = 2;
    return false;
}

// undirected graph
bool dfs(int u, int p = -1) {
    color[u] = 1;
    parent[u] = p;
    for(auto &v : adj[u]) {
        if(v == p) continue;
        if(!color[v]) {
            if(dfs(v, u)) return true;
        }
        else {
            t = u, s = v;
            return true;
        }
    }
    return false;
}
```

3.12 Kuhn - matching

```
vector<int> adj[N]; // [0,n)->[0,m)
int n, m, mat[M];
bool vis[N];
int match(int x){
  if(vis[x]) return 0;
  vis[x] = 1;
  for(auto y : adj[x]) if(mat[y]<0 || match(mat[y])) {mat[y]=x; return 1;}
  return 0;
}
vector<pair<int,int>> max_matching(){
  vector<pair<int, int>> r;
  memset(mat, -1, sizeof(mat));
  for(int i=0; i<n; i++) memset(vis, false, sizeof(vis)), match(i);
  for(int i=0; i<m; i++) if(mat[i] >= 0) r.pb({mat[i], i});
  return r;
}
```

3.13 Hopcroft Karp - matching

```
vector<int> g[MAXN]; // [0,n)->[0,m)
int n,m;
int mt[MAXN],mt2[MAXN],ds[MAXN];
bool bfs(){
  queue<int> q;
  memset(ds,-1,sizeof(ds));
  rep(i,0,n)if(mt2[i]<0)ds[i]=0,q.push(i);
  bool r=false;
  while(!q.empty()){
    int x=q.front();q.pop();
    for(int y:g[x]){
      if(mt[y]>=0&&ds[mt[y]]<0)ds[mt[y]]=ds[x]+1,q.push(mt[y]);
      else if(mt[y]<0)r=true;
    }
  }
  return r;
}
bool dfs(int x){
  for(int y:g[x])if(mt[y]<0||ds[mt[y]]==ds[x]+1&&dfs(mt[y])){
    mt[y]=x;mt2[x]=y;
    return true;
  }
```

```
  ds[x]=1<<30;
  return false;
}
int mm(){
  int r=0;
  memset(mt,-1,sizeof(mt));memset(mt2,-1,sizeof(mt2));
  while(bfs()){
    rep(i,0,n)if(mt2[i]<0)r+=dfs(i);
  }
  return r;
}
```

3.14 Min cost - max flow

```
typedef ll tf;
typedef ll tc;
const tf INFFLOW=1e9;
const tc INFCOST=1e9;
struct MCF{
  int n;
  vector<tc> prio, pot; vector<tf> curflow; vector<int> prevedge,prevnode;
  priority_queue<pair<tc, int>, vector<pair<tc, int>>, greater<pair<tc, int>>> q
      ;
  struct edge{int to, rev; tf f, cap; tc cost;};
  vector<vector<edge>> g;
  MCF(int n):n(n),prio(n),curflow(n),prevedge(n),prevnode(n),pot(n),g(n){}
  void add_edge(int s, int t, tf cap, tc cost) {
    g[s].pb((edge){t,SZ(g[t]),0,cap,cost});
    g[t].pb((edge){s,SZ(g[s])-1,0,0,-cost});
  }
  pair<tf,tc> get_flow(int s, int t) {
    tf flow=0; tc flowcost=0;
    while(1){
      q.push({0, s});
      fill(ALL(prio),INFCOST);
      prio[s]=0; curflow[s]=INFFLOW;
      while(!q.empty()) {
        auto cur=q.top();
        tc d=cur.fst;
        int u=cur.snd;
        q.pop();
        if(d!=prio[u]) continue;
        for(int i=0; i<SZ(g[u]); ++i) {
          edge &e=g[u][i];
          int v=e.to;
          if(e.cap<=e.f) continue;
          tc nprio=prio[u]+e.cost+pot[u]-pot[v];
          if(prio[v]>nprio) {
            prio[v]=nprio;
            q.push({nprio, v});
            prevnode[v]=u; prevedge[v]=i;
            curflow[v]=min(curflow[u], e.cap-e.f);
          }
        }
      }
      if(prio[t]==INFCOST) break;
      rep(i,0,n) pot[i]+=prio[i];
      tf df=min(curflow[t], INFFLOW-flow);
      flow+=df;
      for(int v=t; v!=s; v=prevnode[v]) {
        edge &e=g[prevnode[v]][prevedge[v]];
        e.f+=df; g[v][e.rev].f-=df;
        flowcost+=df*e.cost;
      }
    }
    return {flow,flowcost};
  }
};
```

3.15 2sat

```
struct two_sat {
    int n;
    vector<vector<int>> g, gr;
    vector<int> comp, topological_order, answer;
    vector<bool> vis;

    two_sat(int _n) {
        n = _n;
        g.assign(2 * n, vector<int>());
        gr.assign(2 * n, vector<int>());
        comp.resize(2 * n);
        vis.resize(2 * n);
        answer.resize(2 * n);
    }

    void add_edge(int u, int v) {
        g[u].pb(v);
        gr[v].pb(u);
    }

    void dfs(int u) {
        vis[u] = true;
        each(v, g[u]) if (!vis[v]) dfs(v);
        topological_order.push_back(u);
    }

    void scc(int u, int id) {
```

```
        vis[u] = true, comp[u] = id;
        each(v, gr[u]) if (!vis[v]) scc(v, id);
    }

    bool satisfiable() {
        fill(vis.begin(), vis.end(), false);
        for (int i = 0; i < 2 * n; i++) if (!vis[i]) dfs(i);
        fill(vis.begin(), vis.end(), false);
        reverse(topological_order.begin(), topological_order.end());
        int id = 0;
        for(const auto &v : topological_order) if (!vis[v]) scc(v, id++);
        for(int i=0; i<n; i++) {
            if (comp[i] == comp[i + n]) return false;
            answer[i] = (comp[i] > comp[i + n] ? 1 : 0);
        }
        return true;
    }

    // Conditions
    void add_clause(int a, int b, string op, int c) {
        if(op=="=") {
            if(c==0) add_nor(a, b);
            else if(c==1) add_01_10(a, b);
            else add_and(a, b);
        }
        else if(op=="!=") {
            if(c==0) add_or(a, b);
            else if(c==1) add_same(a, b);
            else add_nand(a, b);
        }
        else if(op=="<") {
            if(c==0) {
                cout<<"No"<<endl;
                exit(0);
            }
            else if(c==1) add_nor(a, b);
            else add_nand(a, b);
        }
        else if(op==">") {
            if(c==0) add_or(a, b);
            else if(c==1) add_and(a, b);
            else {
                cout<<"No"<<endl;
                exit(0);
            }
        }
        else if(op=="<=") {
            if(c==0) add_nor(a, b);
            else if(c==1) add_nand(a, b);
            else return;
        }
        else {
            if(c==0) return;
            else if(c==1) add_or(a, b);
            else add_and(a, b);
        }
    }

    void add_nor(int a, int b) {
        add_edge(a, a+n);
        add_edge(b, b+n);
    }
    void add_01_10(int a, int b){
        add_or(a, b);
        add_nand(a, b);
    }
    void add_and(int a, int b) {
        add_edge(a+n, a);
        add_edge(b+n, b);
    }
    void add_or(int a, int b){
        add_edge(a+n, b);
        add_edge(b+n, a);
    }
    void add_same(int a, int b) {
        add_edge(a, b);
        add_edge(b+n, a+n);
        add_edge(a+n, b+n);
        add_edge(b, a);
    }
    void add_nand(int a, int b) {
        add_edge(a, b+n);
        add_edge(b, a+n);
    }
};

int main() { io
    int n, m; cin>>n>>m;
    two_sat ts(n);
    rep(i,0,m) {
        int a, b, c; string op;
        cin>>a>>b>>op>>c;
        ts.add_clause(a, b, op, c);
    }
    cout<< (ts.satisfiable() ? "Yes" : "No")<<endl;
    return 0;
```

```
}
```

4 Math

4.1 Binpow

```cpp
ll binpow(ll a, ll b, ll m) {
    a %= m;
    ll res = 1;
    while (b > 0) {
        if (b & 1) res = res * a % m;
        a = a * a % m;
        b >>= 1;
    }
    return res;
}
```

4.2 Modular inverse

```cpp
tuple<int, int, int> extendedGCD(int mod, int a) {
  if (!a) return {mod, 1, 0};
  auto[r, x, y] = extendedGCD(a, mod % a);
  return {r, y, x - mod / a * y};
}
int modInverse(int mod, int a) {
    auto[r, x, y] = extendedGCD(mod, a);
    if(r != 1) {
        return -1;
    }
    return ((y < 0) ? y + mod : y);
}
```

4.3 Linear Sieve

```cpp
const int N = 1e7+5; // N: range to get primes
int p[N];
vector<int> pr;
void linearSieve() {
    for(int i=2; i<N; i++) {
        if(!p[i]) {
            p[i] = i;
            pr.push_back(i);
        }
        for(int j=0; i*pr[j] < N; j++) {
            p[i*pr[j]] = pr[j];
            if(pr[j] == p[i])
                break;
        }
    }
}
```

4.4 Sieve of Eratosthenes

```cpp
vector<bool> is_prime(n + 1, true);
is_prime[0] = is_prime[1] = false;
for (int i = 2; i * i <= n; i++) {
    if (is_prime[i]) {
        for (int j = i * i; j <= n; j += i)
            is_prime[j] = false;
    }
}
```

4.5 Greatest Prime Factor

```cpp
vector<int> greatestPrimeFactor(int n){
  vector<int> gp(n+1);
  iota(gp.begin(), gp.end(), 0);
  for(int i = 4; i <= n; i += 2) gp[i] = 2;
  for(int i = 3; i <= n; i += 2)
    if(gp[i] == i)
      for(int j = 2*i; j <= n; j += i)
        gp[j] = i;
  return gp;
}
```

4.6 Smallest Prime Factor

```cpp
vector<int> smallestPrimeFactor(int n) {
    vector<int> spf(n + 1);
    iota(spf.begin(), spf.end(), 0);
    for (int i = 2; i * i <= n; ++i) {
        if (spf[i] == i) {
            for (int j = i * i; j <= n; j += i) {
                if (spf[j] == j) spf[j] = i;
            }
        }
    }
    return spf;
}
```

4.7 Factorials

```cpp
void pre() {
    fact[0] = 1;
    for(int i=1; i<N; i++) {
        fact[i] = 1LL * fact[i-1] * i % MOD;
    }

    ifact[N-1] = binpow(fact[N-1], MOD-2, MOD);
    for(int i=N-2; i>=0; i--) {
        ifact[i] = 1LL * ifact[i+1] * (i+1) % MOD;
    }
```

```
}
```

4.8 Prime factorization

```
set<int> factors;
for(int i=2; i*i<=n; i++) {
    while(!(n % i)) {
        factors.insert(i);
        n /= p;
    }
}
if(n > 1) factors.insert(n);
```

4.9 Divisors

```
vector<int> div;
for(int i=1; i*i<=n; i++) {
    if(!(n % i)) {
        div.pb(d);
        if(d != n/d) div.pb(n / d);
    }
}
sort(all(div));
```

4.10 Identities

$C_n = \frac{2(2n-1)}{n+1} C_{n-1}$
$C_n = \frac{1}{n+1}\binom{2n}{n}$
$C_n \sim \frac{4^n}{n^{3/2}\sqrt{\pi}}$
$\sigma(n) = O(\log(\log(n)))$ (number of divisors of n)
$F_{2n+1} = F_n^2 + F_{n+1}^2$
$F_{2n} = F_{n+1}^2 - F_{n-1}^2$
$\sum_{i=1}^{n} F_i = F_{n+2} - 1$
$F_{n+i}F_{n+j} - F_nF_{n+i+j} = (-1)^n F_iF_j$
(Fermat's little theorem) $a^p \, mod(p) = a \quad a^{p-1} \, mod(p) = 1 \quad a^{p-2} \, mod(p) = a^{-1}$
(Möbius Inv. Formula) Let $g(n) = \sum_{d|n} f(d)$, then $f(n) = \sum d \mid n g(d)\mu\left(\frac{n}{d}\right)$.

5 Strings

5.1 KMP

```
vector<int> prefix_function(const string &s){
    int n=s.length();
    vector<int>pi(n);
    for(int i=1,j=0; i<n; i++){
        while(j>0 && s[i]!=s[j]) j=pi[j-1];
        if(s[i]==s[j])j++;
        pi[i]=j;
    }
    return pi;
}

vector<int> kmp_search(const string &s, const string &t){
    vector<int> ocu, pi=prefix_function(s);
    int n=s.length(), m=t.length();

    for(int i=0,j=0; i<m; i++){
        while(j>0 && t[i]!=s[j]) j=pi[j-1];
        if(t[i]==s[j]) j++;
        if(j==n) {
            ocu.pb(i-j+1);
            j=pi[j-1];
        }
    }
    return ocu;
}
// s=pattern, t=text
```

5.2 Rolling Hash

```
#define rep(i,a,b) for(int i=a;i<b;i++)
using ll = long long;

const int base = 251, MOD = 1e9+7, N = 2e5+5;
ll p[N], hashes[N];
vector<int> rolling_hash(string text, string pattern) {
    int n = text.length(), m = pattern.length();
    p[0] = 1;
    rep(i,1,n+1) p[i] = p[i-1] * base % MOD;

    rep(i,1,n+1) hashes[i] = (hashes[i-1] + (text[i-1]-'a'+1) * p[i-1] % MOD) %
        MOD;

    ll patternHash = 0;
    rep(i,1,m+1) patternHash = (patternHash + (pattern[i-1]-'a'+1) * p[i-1] %
        MOD) % MOD;

    vector<int> ocu;
    rep(i,0,n-m+1) {
        ll curr = (hashes[i+m] - hashes[i] + MOD) % MOD;
        if(curr == patternHash * p[i] % MOD) ocu.push_back(i);
    }
    return ocu;
}

// 0. Define 2 prime numbers: base and mod
// 1. Precompute powers of the base
// 2. Compute prefix hashes for the text
// 3. Compute hash of the pattern
```

```
// 4. Sliding window of pattern size over the text to find matches
```

5.3 LCP using hashing + bs

```
int getLCP(int i, int j, int len) {
    int l = 1, r = len;
    while(l <= r) {
        int m = (l+r) >> 1;
        ll left = (hashes[i+m] - hashes[i] + MOD) % MOD;
        ll right = (hashes[j+m] - hashes[j] + MOD) % MOD;
        left = left * p[j-i] % MOD;
        if(left == right) l = m + 1;
        else r = m - 1;
    }
    return r;
}
// Compute prefix hashes for the text
// LCP: getLCP(idx, idx+len, n)
// length of most repeated substring: LCP+len
```

5.4 Suffix Automaton

```
const int M = 27;
struct state{
  int len, link;
  vector<int> child;
  state(int len = 0, int link = -1): len(len), link(link), child(M, -1){}
  state(int len, int link, const vector<int> &child): len(len), link(link),
      child(child){}
};

struct SuffixAutomaton{
  vector<state> st;
  vector<int> terminals;
  int last = 0;
  SuffixAutomaton(){
    st.emplace_back();
  }
  void extend(char c){
    int curr = st.size();
    st.emplace_back(st[last].len + 1);
    int p = last;
    while(p != -1 && st[p].child[c-'A'] == -1){
      st[p].child[c-'A'] = curr;
      p = st[p].link;
    }
    if(p == -1){
      st[curr].link = 0;
    }
        else{
      int q = st[p].child[c-'A'];
      if(st[p].len + 1 == st[q].len){
        st[curr].link = q;
      }else{
        int clone = st.size();
        st.emplace_back(st[p].len + 1, st[q].link, st[q].child);
        while(p != -1 && st[p].child[c-'A'] == q){
          st[p].child[c-'A'] = clone;
          p = st[p].link;
        }
        st[q].link = st[curr].link = clone;
      }
    }
    last = curr;
  }
  void findTerminals(){
    int p = last;
        while(p > 0) {
            terminals.push_back(p);
            p = st[p].link;
        }
  }
};
// extras
void compute_counts() {
    int n = st.size();
    vector<int> order[n+1];
    for(int i=1; i<n; i++)
        order[st[i].len].pb(i);
    for(int i=n-1; i>=0; i--) {
        for(auto &u : order[i]) {
            int prev = st[u].link;
            if(prev != -1)
                st[prev].cnt += st[u].cnt;
        }
    }
}

ll compute_distinct() {
    ll total = 0;
    for(int i=1; i<(int)st.size(); i++)
        total = total + st[i].len - st[st[i].link].len;
    return total;
}

ll dp_distinct_dfs(int curr) {
    // pre: dp(st)={-1} - init: dfs(0)-1
    auto &x = dp[curr];
```

```
    if(x != -1) return x;
    x = 1;
    for(int i=0; i<M; i++) {
        if(st[curr].child[i] != -1) {
            x += dp_distinct_dfs(st[curr].child[i]);
        }
    }
    return x;
}
// Usage
// SuffixAutomaton sa;
// for(it : s) sa.extend(it);
```

5.5 Suffix Array

```
int sa[N], lcp[N];
int t_sa[N], c[N];
void countingSort (string &s, int ra[], int k) {
  memset(t_sa, 0, sizeof(t_sa));
  memset(c, 0, sizeof(c));
    int sum, n = s.size(), mx = max(300, n);

    for (int i = 0; i < n; i++)
        c[i+k<n?ra[i+k]:0]++;

    for (int i = sum = 0; i < mx; i++) {
        int t = c[i];
        c[i] = sum;
        sum += t;
    }
    for (int i = 0; i < n; i++)
        t_sa[c[sa[i]+k<n?ra[sa[i]+k]:0]++] = sa[i];

    for (int i = 0; i < n; i++) sa[i] = t_sa[i];
}

int t_ra[N], ra[N];
void obtSA (string &s) {
    memset(t_ra, 0, sizeof(t_ra));
    memset(ra, 0, sizeof(ra));
    int r, n = s.size();

    for (int i = 0; i < n; i++) ra[i] = s[i];
    for (int i = 0; i < n; i++) sa[i] = i;
    for (int k = 1; k < n; k <<= 1) {
        countingSort(s, ra, k);
        countingSort(s, ra, 0);
        t_ra[s[0]] = r = 0;
        for (int i = 1; i < n; i++)
            if (ra[sa[i]] == ra[sa[i-1]] && ra[sa[i]+k] == ra[sa[i-1]+k])
                t_ra[sa[i]] = r;
            else
                t_ra[sa[i]] = ++r;
        for (int i = 0; i < n; i++) ra[i] = t_ra[i];
        if (ra[sa[n-1]] == n-1) break;
    }
}

int phi[N], plcp[N];
void obtLCP (string &s) {
    int n = s.size();
    phi[sa[0]] = -1;

    for (int i = 1; i < n; i++) phi[sa[i]] = sa[i-1];
    for (int i = 0, l = 0; i < n; i++) {
        if (phi[i] == -1) {
            plcp[i] = 0;
            continue;
        }
        while (s[i+l] == s[phi[i]+l]) l++;
        plcp[i] = l;
        l = max(l-1, 0);
    }
    for (int i = 0; i < n; i++) lcp[i] = plcp[sa[i]];
}

// Sorting lexicographically using idx l, r
sort(all(res), [](const pair<int, int> &a, const pair<int, int> &b) {
    int szA = a.second-a.first+1, szB = b.second-b.first+1;
    if(a.first == b.first) return szA < szB;
    int prefix = st_query(min(ra[a.first], ra[b.first])+1, max(ra[a.first], ra[b.
        first]));
    int mn = min(szA, szB);
    if(prefix >= mn) {
        if(szA == szB) return a < b;
        return szA < szB;
    }
    return ra[a.first] < ra[b.first];
});
// Count distinct
for(1, n)
    sum += 1ll*n - sa[i] - lcp[i];

// Usage
// s += "#"; obtSA(s); obtLCP(s);
```

5.6 Aho-Corasick

```cpp
struct node {
    int ch[26] = {};
    int fin = 0, suffixLink = 0, endLink = 0;
};
vector<node> aho;

void init() { aho.assign(1, node()); }

void insert(string s) {
    int curr = 0;
    for (int i = 0; i < SZ(s); i++) {
        if (!aho[curr].ch[s[i] - 'a']) {
            aho[curr].ch[s[i] - 'a'] = SZ(aho);
            aho.push_back(node());
        }
        curr = aho[curr].ch[s[i] - 'a'];
    }
    aho[curr].fin = s.size();
}

void initAho() {
    queue<int> q;
    rep(i, 0, 26) if (aho[0].ch[i]) q.push(aho[0].ch[i]);
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        rep(i, 0, 26) {
            int v = aho[u].ch[i];
            if (v) {
                aho[v].suffixLink = aho[aho[u].suffixLink].ch[i], q.push(v);
                if (aho[aho[v].suffixLink].fin) aho[v].endLink = aho[v].suffixLink
                    ;
                else aho[v].endLink = aho[aho[v].suffixLink].endLink;
            }
            else aho[u].ch[i] = aho[aho[u].suffixLink].ch[i];
        }
    }
}

void search(string &t) {
    int u = 0;
    for (int i = 0; i < SZ(t); i++) {
        u = aho[u].ch[t[i] - 'a'];
        for (int v = aho[u].fin ? u : aho[u].endLink; v; v = aho[v].endLink) {
            // patrn de longitud aho[v].fin termina en i
        }
    }
}
```

6 Geometry

6.1 Convex Hull

```cpp
const double eps = 1e-9;
bool leq(double a, double b){return b-a >= -eps;}
bool le(double a, double b){return b-a > eps;}
bool eq(double a, double b){return fabs(a-b) <= eps;}

struct point{
    double x, y;
    int idx = -1;
    point(): x(0), y(0){}
    point(double x, double y): x(x), y(y){}
    point operator-(const point &p) const{return point(x - p.x, y - p.y);}
    point operator*(const int &k) const{return point(x * k, y * k);}
    bool operator<(const point &p) const{return le(x, p.x) || (eq(x, p.x)&& le(y,
        p.y));}
    bool operator==(const point &p) const{return eq(x, p.x) && eq(y, p.y);}
    double cross(const point &p) const{return x * p.y - y * p.x;}
};

istream &operator>>(istream &is, point &p){return is >> p.x >> p.y;}
ostream &operator<<(ostream &os, const point &p){return os << "(" << p.x << ", "
    << p.y << ")";}

vector<point> convexHull(vector<point> P) {
    sort(P.begin(), P.end());
    vector<point> L, U;
    for (int i = 0; i < P.size(); i++) {
        while (L.size() >= 2 && le((L[L.size() - 2] - P[i]).cross(L[L.size() - 1]
            - P[i]), 0)) {
            L.pop_back();
        }
        L.push_back(P[i]);
    }
    for (int i = P.size() - 1; i >= 0; i--) {
        while (U.size() >= 2 && le((U[U.size() - 2] - P[i]).cross(U[U.size() - 1]
            - P[i]), 0)) {
            U.pop_back();
        }
        U.push_back(P[i]);
    }
    L.pop_back();
    U.pop_back();
    L.insert(L.end(), U.begin(), U.end());
    return L;
}
```

```cpp
bool pointInConvexHull(const vector<point> &poly, point p) {
    int n = poly.size();
    if(n < 3) return false;
    rep(i,0,n) {
        point a = poly[i], b = poly[(i+1) % n];
        double cp = (b - a).cross(p - a);
        if(!le(0, cp)) return false;
    }
    return true;
}

int main() {
    ios::sync_with_stdio(0); cin.tie(0);
    int n;
    double h;
    cin >> n >> h;

    vector<point> islands(n);
    vector<double> time(n), heights(n);
    rep(i,0,n) {
        cin >> islands[i] >> heights[i];
        islands[i].idx = i;
        time[i] = max(0.0, h - heights[i]);
    }

    double l = 0.0, r = *max_element(all(time));
    point c;
    cin >> c;
    auto valid = [&](double t) {
        vector<point> aux;
        rep(i,0,n) {
            if(t >= time[i]) aux.pb(islands[i]);
        }
        if(aux.size() < 3) return false;
        aux = convexHull(aux);
        return pointInConvexHull(aux, c);
    };

    if(!valid(r)) {
        cout << -1 << endl;
        return 0;
    }

    for(int it=0; it<100; it++) {
        double m = (l + r) / 2.0;
        if(valid(m)) {
            r = m;
        }
        else {
            l = m;
        }
    }
    cout << fixed << setprecision(12) << r << endl;

    return 0;
}
```

7 Techniques

7.1 MO's algorithm

```cpp
const int N = 3e5+5, B = 550;
int values[N], freq[N], res[N], ans, has[B];
void add(int idx) {
    has[values[idx]/B]++;
    freq[values[idx]]++;
}
void remove(int idx) {
    has[values[idx]/B]--;
    freq[values[idx]]--;
}
int getAns(int k) {
    rep(i,0,B) if(has[i] > k) rep(j,i*B,(i+1)*B)
        if(freq[j] > k) return j;
    return -1;
}
void mos(vector<array<int, 4>> &queries) {
    sort(all(queries), [](array<int, 4> &a, array<int, 4> &b) {
        if(a[0]/B != b[0]/B) return a[0] < b[0];
        return (a[0]/B & 1 ? a[1] < b[1] : a[1] > b[1]);
    });

    int l = 0, r = -1;
    for(auto &[ql, qr, k, idx] : queries) {
        while(l > ql) add(--l);
        while(r < qr) add(++r);
        while(l < ql) remove(l++);
        while(r > qr) remove(r--);
        res[idx] = getAns(k);
    }
}
```

7.2 Parallel binary search

```cpp
void parallel_bs() {
    vector<array<int, 2>> queries(n+1);
    rep(i,1,n+1) queries[i] = {1, q};

```

```
    bool f = 1;
    while(f) {
        f = 0;
        vector<vector<int>> mids(m+2);
        for(auto &it : queries) {
            if(it.l <= it.r) {
                int mid = (it.l + it.r) >> 1;
                mids[mid].pb(it.idx);
                f = 1;
            }
        }

        obj.reset();

        rep(mid,1,q+1) {
            auto &[l, r, k] = values[mid];
            obj.update(l, r, k);

            for(auto &idx : mids[mid]) {
                Queries &q = queries[idx];
                if(ok) q.r = mid - 1;
                else q.l = mid + 1;
            }
        }
    }
}
```

7.3 Split objects into light and heavy

```
vector<pair<int, int>> adj[N];
bitset<N> heavy;
const int B = 2050;
vector<pair<int, int>> heavyVertices[N];
int values[N], deg[N];

void solve() {
    ll res = 0;
    auto dfs = [&](auto self, int u, int p = -1) -> void {
        for(auto &[v, x] : adj[u]) {
            if(v == p) continue;
            if(values[u] != values[v]) res += 1ll * x;
            deg[u]++, deg[v]++;
            self(self, v, u);
        }
    };
    dfs(dfs, 0);

    rep(i,0,n) if(deg[i] >= B) heavy[i] = 1;

    rep(u,0,n) {
        if(heavy[u]) {
            for(auto &[v, x] : adj[u]) {
                colors[u][values[v]] += 1ll * x;
                if(heavy[v])
                    heavyVertices[u].pb({v, x});
                    heavyVertices[v].pb({u, x});
            }
        }
    }

    while(q--) {
        int u, c;
        cin >> u >> c;
        u--;

        if(heavy[u]) {
            res += 1ll * colors[u][values[u]];
            res -= 1ll * colors[u][c];
            for(auto &[v, x] : heavyVertices[u]) {
                colors[v][values[u]] -= 1ll * x;
                colors[v][c] += 1ll * x;
            }
            values[u] = c;
        }
        else {
            for(auto &[v, x] : adj[u]) {
                if(values[u] != values[v]) {
                    if(c == values[v]) res -= 1ll * x;
                }
                else {
                    if(c != values[v]) {
                        res += 1ll * x;
                    }
                }
                if(heavy[v]) {
                    colors[v][values[u]] -= 1ll * x;
                    colors[v][c] += 1ll * x;
                }
            }
            values[u] = c;
        }

        cout << res << endl;
    }
}
```

7.4 Meet in the middle

```cpp
int main() {
    ios::sync_with_stdio(0); cout.tie(0); cin.tie(0);
    int n; cin >> n;
    vector<long long> a(n), b(n);
    for(auto &it : a) cin >> it;
    for(auto &it : b) cin >> it;

    auto subsets = [](vector<long long> &a, vector<long long> &b, bool f){
        int n = a.size();
        vector<long long> aux;
        for(int mask=0; mask<(1<<n); mask++) {
            long long sum = 0;
            for(int j=0; j<n; j++) {
                if((1 << j) & mask) sum += a[j];
                else sum -= b[j];
            }
            if(f) sum = abs(sum);
            aux.push_back(sum);
        }
        return aux;
    };

    if(n <= 20) {
        vector<long long> res = subsets(a, b, 1);
        sort(res.begin(), res.end());
        for(auto &it : res) cout << it << " ";
        cout<< endl;
        return 0;
    }

    vector<long long> leftA(a.begin(), a.begin()+n/2),
                      leftB(b.begin(), b.begin()+n/2),
                      rightA(a.begin()+n/2, a.end()),
                      rightB(b.begin()+n/2, b.end());

    vector<long long> left = subsets(leftA, leftB, 0),
                      right = subsets(rightA, rightB, 0);
    sort(right.begin(), right.end());

    priority_queue<tuple<long long, int, int, int>, vector<tuple<long long, int,
         int, int>>, greater<tuple<long long, int, int, int>>> pq;
    for(int i=0; i<left.size(); i++) {
        int idx = lower_bound(right.begin(), right.end(), -left[i]) - right.begin
            ();
        long long best = LONG_LONG_MAX, curr;
        int idxBest;
        for(int j=-1; j<=1; j++) {
            if(idx + j >= 0 && idx + j < right.size()) {
                curr = abs(left[i] + right[idx+j]);
                if(curr < best) {
                    best = curr;
                    idxBest = idx+j;
                }
            }
        }
        pq.push({best, idxBest, i, 0});
    }

    int req = 1 << 20, cnt = 0;
    while(!pq.empty() && cnt < req) {
        auto [sum, idx, i, event] = pq.top();
        pq.pop();
        cout << sum << " ";
        cnt++;
        if(event <= 0 && idx-1 >= 0) {
            long long lsum = abs(left[i] + right[idx-1]);
            pq.push({lsum, idx-1, i, -1});
        }
        if(event >=0 && idx+1 < right.size()) {
            long long rsum = abs(left[i] + right[idx+1]);
            pq.push({rsum, idx+1, i, 1});
        }
    }
    cout << endl;

    return 0;
}
```

7.5 Dijkstra on ST

```cpp
vector<pair<int, int>> adj[2*4*N];
ll dist[2*4*N];
int mp[N], vis[2*4*N];
void build(int l, int r, int i) {
    if (l == r) {
        mp[l] = i;
        adj[i].pb({i+4*N, 0});
        adj[i+4*N].pb({i, 0});
        return;
    }
    int m = (l+r)>>1;
    build(l, m, 2*i+1);
    build(m+1, r, 2*i+2);
    adj[i].pb({2*i+1, 0});
    adj[i].pb({2*i+2, 0});
    adj[2*i+1+4*N].pb({i+4*N, 0});
    adj[2*i+2+4*N].pb({i+4*N, 0});
```

```
}
void add(int l, int r, int u, int a, int b, int w, int op, int i) {
    if (a > r || b < l) return;
    if (a <= l && r <= b) {
        if(op == 2) adj[mp[u]].pb({i, w});
        else adj[i+4*N].pb({mp[u]+4*N, w});
        return;
    }
    int m = l+r>>1;
    add(l, m, u, a, b, w, op, 2*i+1);
    add(m+1, r, u, a, b, w, op, 2*i+2);
}
```

7.6 Venice set

```
struct VeniceSet {
    multiset<int> st;
    int global = 0;

    void add(int x) {
        st.insert(x + global);
    }
    void remove(int x) {
        st.erase(st.find(x + global));
    }
    void updateAll(int x) {
        global += x;
    }
    int min(int x) {
        return *st.begin() - global;
    }
};
```

7.7 Divide and conquer DP

```
// Usar cuando: dp[i][j] = min_{k<j} dp[i-1][k] + cost(k+1, j),
// cost se calcula facil en O(1) u O(log n), y los grupos no se
// tocan entre si y son continuos (opt[i][j] <= opt[i][j+1]).
// dp[i][mid]: el i-esimo grupo tiene que acabar exactamente en mid.
// O(K * N log N * cost). Llamar compute(i, 1, n, 0, n-1) para i = 1..K
ll dp[K][N];
// costo de un grupo que empieza en l y acaba en r, O(1) u O(log n)
ll cost(int l, int r) {
    // ...
}
void compute(int i, int l, int r, int optl, int optr) {
    if(l>r)return;
    int mid=(l + r)/2;
    ll best_val=1e17;
    int best_k = optl;
    for (int k = optl; k <= min(mid, optr); ++k) {
        ll val = dp[i-1][k]+cost(k+1,mid);
        if (val<best_val) {
            best_val=val;
            best_k=k;
        }
    }
    dp[i][mid]=best_val;
    compute(i,l,mid-1,optl,best_k);
    compute(i,mid+1, r,best_k,optr);
}
```

7.8 SOS DP

```
// SOS DP (Sum Over Subsets). O(N * 2^N)
// N el numero maximo de bits y M = 2^N - 1 el maximo numero
const ll N=20,M=(1<<N)-1;
ll reps[M+1];
// Antes: reps[x] = valor de x (p. ej. cuantas veces aparece x)
// Despues: reps[x] = suma de reps[y] para todo y submascara de x (y & x == y)
// Se tiene que mirar como un trie para entender las secuencias o cambiarlas
void dp(){
    for (int i=0; i<N; i++) {
        for (int x=0; x<=M; x++) {
            if (x & (1 << i)) { reps[x] += reps[x ^ (1 << i)]; }
        }
    }
    // La respuesta esta en reps[x]
}
```

7.9 Boolean DP with bitset

```
// DP booleana con bitset
// (subset sum / mochila 0-1 de alcanzabilidad)
// O(n * N / 64)
// N la maxima suma alcanzable (el maximo de una pila)
const int N=1e5;
// Los numeros o las transiciones que tengo
vector<int> nums;
bitset<N> dp(){
    // Antes era bool dp[pos][cant]
    bitset<N> dp;
    // Equivale a no poner nada
    dp[0]=1;
    // Cada numero se pone o no se pone dentro de la pila
    for(auto it:nums)dp|=dp<<it;
    // dp[s]=1 si la suma s se puede formar
    return dp;
}
```

8 Basic techniques

8.1 Difference array

```
// If we only need to get the final result, we can increase l and r+1
// to simulate range update and use a prefix sum to get the answer.
diff[l] += k;
diff[r+1] -= k;

// If we have initial values, we need to subtract the previous values:
diff[i] -= values[i - 1];
```

8.2 Prefix Sum 2D

```
// Compute prefix sum 2D
rep(i,1,n+1) rep(j,1,m+1)
    prefix[i][j] = prefix[i-1][j] + prefix[i][j-1]
                                  - prefix[i-1][j-1]
                                  + matrix[i-1][j-1]

// Query prefix sum 2D
query = prefix[i][j] - prefix[i-1][j] - prefix[i][j-1] + prefix[i-1][j-1]

// Specific parts
query = prefix[r2][c2] - prefix[r1-1][c2] - prefix[r2][c1-1] + prefix[r1-1][c1
    -1];
```

8.3 Random number generator

```
// Secure random seed
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());

// Generate values using distribution
uniform_int_distribution<int> dist(0, N);  // [0, N]
randval = dist(rng);
```

8.4 Coordinate compression

```
// Push to an array all possible values
coord.pb(x);
sort(all(coord));
coord.resize(unique(all(coord)) - coord.begin());

auto get = [&](int x) {
    return lower_bound(all(coord), x) - coord.begin();
};
```

8.5 Digit DP

```
int dp[11][2][2];
auto calc = [&](auto self, int idx, bool f1, bool f2) -> int {
    if(idx >= n) return 0;

    int &x = dp[idx][f1][f2];
    if(x != -1) return x;

    int num1 = a[idx] - '0', num2 = b[idx] - '0';
    int l = (f1 ? 0 : num1);
    int r = (f2 ? 9 : num2);
    int mn = 1e9+5;

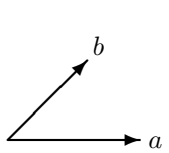
    rep(i,l,r+1) {
        mn = min(mn, self(self, idx+1, f1 | (i > num1), f2 | (i < num2)) + (num1
            ==i) + (num2==i));
    }
    return x = mn;
};
memset(dp, -1, sizeof(dp));
cout << calc(calc, 0, 0, 0) << endl;
```

8.6 Intersection [L1, R1] and [L2, R2]

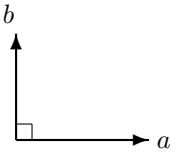
```
We need to check whether there is any number common to both ranges:
max(L1, L2) <= min(R1, R2)
```

9 Notas & fórmulas

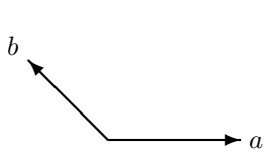
9.1 Producto punto



$a \cdot b > 0$
b en la misma dirección que a



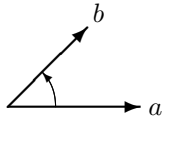
$a \cdot b = 0$
b perpendicular a a



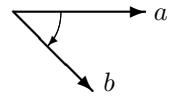
$a \cdot b < 0$
b en sentido contrario a a

$$a \cdot b = a_x b_x + a_y b_y$$

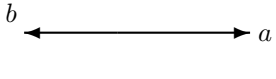
9.2 Producto cruz



$a \times b > 0$
b a la izquierda de a



$a \times b < 0$
b a la derecha de a



$a \times b = 0$
b colineal con a

$$a \times b = a_x b_y - a_y b_x$$

9.3 Leyes de los exponentes

$$a^m \cdot a^n = a^{m+n} \qquad \frac{a^m}{a^n} = a^{m-n} \qquad (a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n \qquad \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n} \qquad a^0 = 1$$

$$a^{-n} = \frac{1}{a^n} \qquad a^{\frac{1}{n}} = \sqrt[n]{a} \qquad a^{\frac{m}{n}} = \sqrt[n]{a^m}$$

9.4 Logaritmos

$$\log_b x = y \iff b^y = x \qquad \log_b 1 = 0 \qquad \log_b b = 1$$

$$\log_b(xy) = \log_b x + \log_b y \qquad \log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y \qquad \log_b(x^n) = n \log_b x$$

$$b^{\log_b x} = x \qquad \log_b x = \frac{\ln x}{\ln b} \qquad a^{\log_b c} = c^{\log_b a}$$

9.5 Integrales

$$\int c\,dx = cx \qquad \int x^n\,dx = \frac{x^{n+1}}{n+1} \qquad \int \frac{1}{x}\,dx = \ln|x|$$

$$\int e^x\,dx = e^x \qquad \int a^x\,dx = \frac{a^x}{\ln a} \qquad \int \ln x\,dx = x\ln x - x$$

$$\int \sin x\,dx = -\cos x \qquad \int \cos x\,dx = \sin x \qquad \int \sec^2 x\,dx = \tan x$$

$$\int \frac{dx}{1+x^2} = \arctan x \qquad \int \frac{dx}{\sqrt{1-x^2}} = \arcsin x \qquad \int cf = c\int f$$

$$\int (f \pm g) = \int f \pm \int g \qquad \int u\,dv = uv - \int v\,du \qquad \int_a^b f(x)\,dx = F(b) - F(a)$$

9.6 Linealidad del valor esperado

Siempre	$E[X+Y] = E[X] + E[Y]$
(dependientes o	$E[aX+b] = a\,E[X] + b$
independientes)	$E\left[\sum X_i\right] = \sum E[X_i]$
Independientes	$E[XY] = E[X]\,E[Y]$
	$Var(X+Y) = Var(X) + Var(Y)$
Dependientes	$E[XY] = E[X]\,E[Y] + Cov(X,Y)$
	$Var(X+Y) = Var(X) + Var(Y) + 2\,Cov(X,Y)$

9.7 Teorema de Bayes

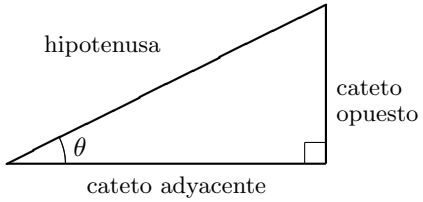
$$P(A \mid B) = \frac{P(B \mid A)\,P(A)}{P(B)} \qquad P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$$

Probabilidad total: $P(B) = \sum_i P(B \mid A_i)\,P(A_i)$ (A_i disjuntos que cubren todo)

9.8 Probabilidad discreta y continua

Discreta	**Continua**
$P(X = x) = p(x)$	$P(a \le X \le b) = \int_a^b f(x)\,dx$
$\sum_x p(x) = 1$	$\int_{-\infty}^{\infty} f(x)\,dx = 1$
$E[X] = \sum_x x\,p(x)$	$E[X] = \int_{-\infty}^{\infty} x\,f(x)\,dx$
$Var(X) = E[X^2] - E[X]^2$	

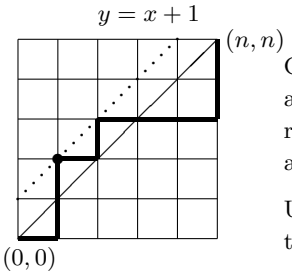
9.9 Seno, coseno y tangente



$$\sin\theta = \frac{\text{opuesto}}{\text{hipotenusa}} \qquad \cos\theta = \frac{\text{adyacente}}{\text{hipotenusa}} \qquad \tan\theta = \frac{\text{opuesto}}{\text{adyacente}}$$

9.10 Números de Catalan (principio de reflexión)

$$C_n = \binom{2n}{n} - \binom{2n}{n+1} = \frac{1}{n+1}\binom{2n}{n} \qquad C_{n+1} = \sum_{i=0}^{n} C_i\,C_{n-i}$$



Caminos de $(0,0)$ a (n,n) (derecha/arriba) sin pasar arriba de $y = x$. Un camino malo toca $y = x+1$: se refleja lo que sigue después del primer toque y acaba en $(n-1, n+1)$ $\Rightarrow$ malos $= \binom{2n}{n+1}$.

Usos: paréntesis balanceados, árboles binarios, triangulaciones, caminos de Dyck.

9.11 Teorema de Lucas

$$\binom{n}{m} \equiv \prod_i \binom{n_i}{m_i} \pmod{p} \qquad (n_i,\ m_i\text{: dígitos de } n,\ m \text{ en base } p,\ p \text{ primo})$$

Si algún $m_i > n_i$ el resultado es 0. $\binom{n}{m}$ es impar $\iff$ `(n & m) == m`

9.12 Funciones multiplicativas

Multiplicativa: $f(ab) = f(a)\,f(b)$ si $\gcd(a,b) = 1$

Completamente multiplicativa: $f(ab) = f(a)\,f(b)$ siempre

Se separa en primos: $f(p_1^{e_1} \cdots p_k^{e_k}) = f(p_1^{e_1}) \cdots f(p_k^{e_k})$

Solo hay que saber $f(p^e)$, y se calcula para todo $n \le N$ con la criba lineal.

Función	Fórmula	Para qué sirve
$\varphi(n)$ (Euler)	$n \prod_{p\mid n} \left(1 - \frac{1}{p}\right)$	Contar $i \le n$ con $\gcd(i,n) = 1$. Euler: $a^{\varphi(m)} \equiv 1 \pmod m$
$\sigma_0(n)$	$\prod (e_i + 1)$	Número de divisores de n
$\sigma_1(n)$	$\prod \frac{p_i^{e_i+1}-1}{p_i-1}$	Suma de los divisores de n
$\mu(n)$ (Möbius)	0 si $p^2 \mid n$, si no $(-1)^r$	Inclusión-exclusión sobre divisores: $[\gcd = 1] = \sum_{d\mid \gcd} \mu(d)$
$\gcd(n,m)$, m fijo	$\prod p_i^{\min(e_i,f_i)}$	Sumas de gcd: $\sum_{i\le n} \gcd(i,n) = \sum_{d\mid n} d\,\varphi(\frac{n}{d})$
n^k	completamente mult.	Base para armar otras con convolución

$$n = \prod p_i^{e_i}, \quad m = \prod p_i^{f_i}, \quad r = \text{número de primos distintos de } n$$
$$\sum_{d\mid n} \varphi(d) = n \qquad \sum_{d\mid n} \mu(d) = [n = 1]$$

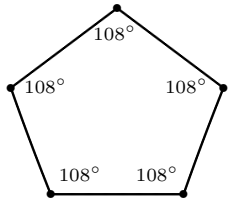
9.13 Fórmula del zapato (área de polígonos)

$$A = \frac{1}{2} \left| \sum_{i=0}^{n-1} p_i \times p_{(i+1)\,\%\,n} \right|$$

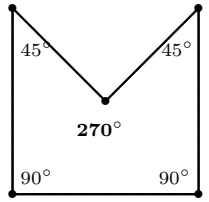
Triángulo con signo (sin valor absoluto): > 0 si a, b, c van en sentido antihorario, < 0 si horario, $= 0$ si son colineales.

$$2A = (b-a) \times (c-a) = x_a(y_b - y_c) + x_b(y_c - y_a) + x_c(y_a - y_b)$$

9.14 Polígono convexo y no convexo



Convexo
todos los ángulos $< 180°$



No convexo
algún ángulo $> 180°$

9.15 Bitwise tricks

```
x & (1<<i)          // checar el bit i
x | (1<<i)          // prender el bit i
x & ~(1<<i)         // apagar el bit i
x ^ (1<<i)          // invertir el bit i
~x == -x-1          // complemento (invierte todos los bits)
x ^ ((1<<n)-1)      // complemento solo de los n bits bajos
x & -x              // bit menos significativo prendido (lowbit)
x & (x-1)           // apagar el bit menos significativo
(x & (x-1)) == 0    // x es potencia de 2 (o x == 0)
x | (x+1)           // prender el 0 menos significativo
(1<<n) - 1          // n bits prendidos (usar 1LL si n >= 31)
a + b == (a ^ b) + 2*(a & b)
a + b == (a | b) + (a & b)
a ^ b ^ b == a      // xor es su propio inverso
// xor de 0..n: n%4==0 -> n, 1 -> 1, 2 -> n+1, 3 -> 0
__builtin_popcountll(x) // cantidad de bits prendidos
__builtin_ctzll(x)      // ceros al final (x != 0)
__builtin_clzll(x)      // ceros al inicio (x != 0)
63 - __builtin_clzll(x) // floor(log2(x)), indice del bit mas alto
__builtin_parityll(x)  // popcount % 2
// Iterar submascaras de m en orden decreciente, O(3^n) en total
for (int s = m; s > 0; s = (s-1) & m) { /* ... */ } // s = 0 aparte
// Iterar supermascaras de m con n bits
for (int s = m; s < (1<<n); s = (s+1) | m) { /* ... */ }
// Siguiente numero con la misma cantidad de bits (Gosper)
ll c = x & -x, r = x + c; x = (((r ^ x) >> 2) / c) | r;
```

10 Funciones de Python y C++

10.1 Python - Strings

```
s = "Hola Mundo"
s.lower(), s.upper(), s.swapcase(), s.title(), s.capitalize()
s.isdigit(), s.isalpha(), s.isalnum(), s.isspace(), s.islower(), s.isupper()
s.find("o"), s.rfind("o")   # indice o -1
```

```python
s.index("o")                # como find pero lanza ValueError
s.count("o")                # ocurrencias sin traslape
s.startswith("Ho"), s.endswith("do")
s.replace("o", "0")         # replace(a, b, k): solo las primeras k
s.split()                   # por espacios, ignora espacios repetidos
"a,b,,c".split(",")         # ['a', 'b', '', 'c']
s.rsplit(" ", 1)            # corta desde la derecha, max 1 vez
s.partition(" ")            # ('Hola', ' ', 'Mundo')
" x ".strip(), " x ".lstrip(), " x ".rstrip() # strip("ab") quita esos chars
" ".join(["a", "b", "c"])   # 'a b c'
s[::-1]                     # reverso
s[2:5], s[-3:]              # slicing [l, r)
"7".zfill(3), "ab".ljust(4, "."), "ab".rjust(4), "ab".center(6, "*")
ord("a"), chr(97)           # 97, 'a'
int("ff", 16), int("101", 2) # 255, 5 (string en base b a int)
bin(5)[2:], hex(255)[2:], oct(8)[2:], format(5, "08b") # '101', 'ff', '10',
    '00000101'
f"{3.14159:.2f}", f"{42:>5}", f"{7:03d}"           # '3.14', '  42', '007'
sorted(s), "".join(sorted(s)) # lista de chars ordenada
list(s), set(s)
from collections import Counter
Counter("banana").most_common(1) # [('a', 3)]
import string
string.ascii_lowercase, string.ascii_uppercase, string.digits
s.translate(str.maketrans("aeiou", "AEIOU")) # reemplazo char a char
# Leer rapido toda la entrada
import sys
data = sys.stdin.buffer.read().split() # tokens en bytes, usar int(x)
```

10.2 C++ - Strings (stoi, to_string, split)

```cpp
// string -> numero (lanzan excepcion si no empieza con numero)
int a = stoi("123");       // stoi("ff", nullptr, 16) en base 16
ll b = stoll("123456789012");
double c = stod("3.14");
// numero -> string
string s = to_string(42);  // to_string(3.5) da "3.500000"
// Separar por un delimitador
vector<string> split(const string& s, char d = ' ') {
    vector<string> res;
    string tok;
    stringstream ss(s);
    while (getline(ss, tok, d)) res.push_back(tok); // "a,,b" -> {"a","","b"}
    return res;
}
// Version ll: separa y convierte cada token
vector<ll> splitll(const string& s, char d = ' ') {
    vector<ll> res;
    for (auto& t : split(s, d)) if (!t.empty()) res.push_back(stoll(t));
    return res;
}
// Leer una linea completa despues de cin >> x: cin.ignore() o cin >> ws
// getline(cin >> ws, linea);
```

10.3 C++ - __int128

```cpp
// __int128 llega a ~1.7e38. No se puede usar con cin/cout ni con literales > ll
// Ojo: getchar/putchar no mezclan bien con cin/cout si se usa sync_with_stdio(
    false)
__int128 read() {
    __int128 x = 0, f = 1;
    char ch = getchar();
    while (ch < '0' || ch > '9') {
        if (ch == '-') f = -1;
        ch = getchar();
    }
    while (ch >= '0' && ch <= '9') {
        x = x * 10 + ch - '0';
        ch = getchar();
    }
    return x * f;
}
void print(__int128 x) {
    if (x < 0) {
        putchar('-');
        x = -x;
    }
    if (x > 9) print(x / 10);
    putchar(x % 10 + '0');
}
// Para cout: convertir a string
string to_str(__int128 x) {
    if (x == 0) return "0";
    bool neg = x < 0;
    string s;
    while (x) { s += char('0' + (neg ? -(x % 10) : x % 10)); x /= 10; }
    if (neg) s += '-';
    reverse(s.begin(), s.end());
    return s;
}
bool cmp(__int128 x, __int128 y) { return x > y; }
// Multiplicar dos ll sin overflow: (__int128)a * b % mod
```